Determine the value of each digit

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: What is the value of 3 in 8,364,219? Copy or complete the first step.

2. Partition 5,082,407.

3. What is the value of 7 in 4,271,650?

4. Miro says: Miro says the 8 in 5,082,407 is worth 8,000. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: What is the value of 3 in 8,364,219? Copy or complete the first step.
   Answer 300,000.

2. Partition 5,082,407.
   Answer 5,000,000 + 80,000 + 2,000 + 400 + 7.

3. What is the value of 7 in 4,271,650?
   Answer 70,000.

4. Miro says: Miro says the 8 in 5,082,407 is worth 8,000. Is this correct? Explain.
   Answer The 8 sits in the ten-thousands column, so it is worth 80,000.

Determine the value of each digit

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. What is the value of 3 in 8,364,219?

2. Partition 5,082,407.

3. What is the value of 7 in 4,271,650?

4. Solve this independently and show every stage: What is the value of 3 in 8,364,219?

5. Use a second method or representation for: Partition 5,082,407.

6. Check this result using an inverse, estimate or known fact: What is the value of 7 in 4,271,650?

7. A number has 6 millions, 4 ten-thousands, 9 hundreds and 2 ones. What is it?

8. Identify and correct this misconception: Miro says the 8 in 5,082,407 is worth 8,000.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: What is the value of 7 in 4,271,650?

Teacher answers and checking prompts

1. What is the value of 3 in 8,364,219?
   Answer 300,000.

2. Partition 5,082,407.
   Answer 5,000,000 + 80,000 + 2,000 + 400 + 7.

3. What is the value of 7 in 4,271,650?
   Answer 70,000.

4. Solve this independently and show every stage: What is the value of 3 in 8,364,219?
   Answer 300,000.

5. Use a second method or representation for: Partition 5,082,407.
   Answer Expected result: 5,000,000 + 80,000 + 2,000 + 400 + 7. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: What is the value of 7 in 4,271,650?
   Answer Expected result: 70,000. A valid check must be shown.

7. A number has 6 millions, 4 ten-thousands, 9 hundreds and 2 ones. What is it?
   Answer 6,040,902.

8. Identify and correct this misconception: Miro says the 8 in 5,082,407 is worth 8,000.
   Answer The 8 sits in the ten-thousands column, so it is worth 80,000.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: What is the value of 7 in 4,271,650?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Determine the value of each digit

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. A number has 6 millions, 4 ten-thousands, 9 hundreds and 2 ones. What is it?

2. Prove the answer to this question, rather than only calculating it: What is the value of 3 in 8,364,219?

3. Solve this in two different ways and compare them: Partition 5,082,407.

4. Work backwards from the answer to reconstruct the question: 70,000.

5. Explain why this reasoning fails: Miro says the 8 in 5,082,407 is worth 8,000.

6. Create a new example that tests the same idea as: What is the value of 7 in 4,271,650?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. A number has 6 millions, 4 ten-thousands, 9 hundreds and 2 ones. What is it?
   Answer 6,040,902.

2. Prove the answer to this question, rather than only calculating it: What is the value of 3 in 8,364,219?
   Answer Expected result: 300,000. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Partition 5,082,407.
   Answer Expected result: 5,000,000 + 80,000 + 2,000 + 400 + 7. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 70,000.
   Answer One valid reconstruction is: What is the value of 7 in 4,271,650? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro says the 8 in 5,082,407 is worth 8,000.
   Answer The 8 sits in the ten-thousands column, so it is worth 80,000.

6. Create a new example that tests the same idea as: What is the value of 7 in 4,271,650?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.